Subin Kim

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Education

University of Pennsylvania **Pennsylvania, U.S.**

Doctor of Philosophy, Electrical and Systems Engineering *Aug. 2025 –*

- Advisor: Dinesh Jayaraman

Korea Advanced Institute of Science & Technology (KAIST) **Daejeon, South Korea**

Master of Science, School of Computing *Mar. 2023 – Feb. 2025*

- Advisor: Sungjin Ahn

Bachelor of Science, School of Computing *Mar. 2018 – Feb. 2023*

- Minor in Industrial & Systems Engineering

Publications

*: equal contributions

[1] {**Subin Kim***, Qinhe Peng*}, Dinesh Jayaraman, "Words Are Not Motions: Steering Vision-Language-Action Models with Jacobian Pullbacks", *SPAIS WS @ CoRL 2026, Awarded Robocurve Travel Grant (Top 7.6%).*

[2] {Miranda Muqing Miao*, **Subin Kim***}, Brandon Yang, Lyle Ungar, "Contrastive Conceptor Activation Steering (COAST): Unlocking Vision-Language-Action Models through Hidden States", *Mechanistic Interpretability WS @ ICML 2026.*

[3] **Subin Kim**, Anh-Quan Pham, Dinesh Jayaraman, " Zero-Shot VLA Adaptation with Retrieved Guidance", *Under Review, 2026.*

[4] {Arnav Kumar* Jain, Vibhakar Mohta*}, **Subin Kim**, Atiksh Bhardwaj, Juntao Ren, Yunhai Feng, Sanjiban Choudhury, Gokul Swamy, "A Smooth Sea Never Made a Skilled SAILOR:Robust Imitation via Learning to Search", *NeurIPS 2025 spotlight.*

[5] {Hany Hamed*, **Subin Kim***}, Dongyeong Kim, Jaesik Yoon, Sungjin Ahn, "Dr. Strategy: Model-Based Generalist Agents with Strategic Dreaming", *ICML, 2024.*

[6] **Subin Kim**, Prin Phunyaphibarn, Donghyun Ahn, and Sundong Kim, "Playgrounds for Abstraction and Reasoning", *NeurIPS Workshop for on neuro Causal and Symbolic AI* **(NeurIPS nCSIW)**, *2022.* *Korea Software Congress* (KSC), *2022* **(Best Paper Award)**.

[7] **Subin Kim**, Prin Phunyaphibarn, Donghyun Ahn, Sundong Kim, "Test Bed for Abstraction and Reasoning", *Journal of KIISE, Vol. 51, No. 1, 2024.*

[8] **Subin Kim**, Sumin Han, Dongman Lee, "Analysis of impact on Seoul traffic congestion by Land Use using travel speed and POI", *Korea Software Congress(KSC), 2021.*

Teaching Experience

Korea Advanced Institute of Science & Technology (KAIST) **Daejeon, South Korea**

TA, Deep Reinforcement Learning and Game AI (CS492) **(Best TA Award)** Mar. 2024 – June. 2024

TA, Hana Bank-KAIST Digital Warrior Program Oct. 2021 – June. 2022

TA, Korean for International Students (HSS193, HSS586) Mar. 2022 – Dec. 2022

TA, Korean Camp for International Students Jan. 2022 - Feb. 2022

Awards & Scholarships

Robocurve Travel Grant, SPAIS @ CoRL 2026 — 2026
National Scholarship, KAIST (Aided by Government of South Korea) — 2023 – 2025
Best TA Award, Deep Reinforcement Learning and Game AI (CS492) — 2024
Best Paper Award, Korea Software Congress (KSC) — 2022
National Excellence Scholarship, Korea Student Aid Foundation (KOSAF) — 2019 – 2022
National Scholarship, Korea Student Aid Foundation (KOSAF) — 2018

Miscellaneous

Google ExploreCSR — **Daejeon, South Korea**
Workshop Organizer — Sep. 2023 - Aug. 2024

KAIST swimming team KAORI — **Daejeon, South Korea**
Certified Korean Red Cross lifeguard — Jan. 2023 – Present
Senior member — Jun. 2021 - Present
50m Silver medal in women backstroke (Daejeon Masters) — Jun. 2022
50m Bronze medal in women freestyle (Daejeon Masters) — Jun. 2022